HERIOT WATT UNIVERSITY

SCHOOL OF MATHEMATICAL AND COMPUTER SCIENCES

Department of Actuarial Mathematics & Statistics

F78AA

ACTUARIAL & FINANCIAL MATHEMATICS A

Semester 1 2021/22

Duration: Two Hours

Total marks: 60

ANSWER ALL SIX QUESTIONS

A UNIVERSITY APPROVED CALCULATOR MAY BE USED

1. Give your answers rounded to 6 decimal places in the following questions.

(a) Given that the effective rate of discount is 0.12% per month, calculate the effective rate of interest per annum.

(2 marks)

(b) Given that the effective rate of interest is 0.21% per month. Calculate the effective rate of discount per half-year.

(2 marks)

(Total 4 marks)

2. Daniel purchases a perpetuity with varying annual payments made in arrears. During the first 5 years, the payments are constant and equal to 10 per year. Beginning in year 6, the payments start to increase. For year 6 and all subsequent years, the payment in that year is p% larger than the payment in the year immediately preceding that year, where $p < 9.2$.

At an annual effective interest rate of 9.2%, the perpetuity has a present value of 167.50. Calculate p, to 2 decimal places.

(Total 6 marks)

3. Susan deposits \$2000 into a bank account that pays interest at the following rates:

- an effective rate of interest of 4% per annum for the first three months,
- an effective rate of interest of 2.5% per annum for the following eight months, and
- an effective rate of interest of 3% per annum thereafter.

(a) Calculate, correct to 2 decimal places, the accumulated value of Susan's deposit after exactly 6 months have elapsed.

(3 marks)

(b) Calculate the least integer number of months she must leave that \$2000 in the account for it to grow to at least \$2200.

(5 marks)

(Total 8 marks)

PLEASE TURN OVER

4. Albert purchased a bond with exactly 20 years to redemption. The bond pays annual coupons, in arrears, of 5% per annum and is redeemed at par. Albert, who is not liable to pay tax, will obtain a gross redemption yield of 6% per annum if he holds the bond utill redemption.

(a) Calculate the purchase price Albert paid for the bond, per $100 nominal if his intention was to hold the bond utill redemption.

(3 marks)

(b) After exactly ten years, immediately after receiving payment of the coupon then due, Albert sells the bond to Vicky who is liable to pay income tax and capital gains tax at a rate of 30%. The bond is purchased by Vicky to provide a net rate of return of 6.5% per annum.

(i) Calculate the price Vicky paid for the bond, per $100 nominal.

(5 marks)

(ii) Using trial-and-error followed by linear interpolation, calculate the annual effective rate of return, correct to 4 decimal places, earned by Albert during the period for which he held the bond.

(5 marks)

(Total 13 marks)

5. A loan is repayable by a decreasing annuity payable annually in arrears for 20 years. The repayment at the end of the first year is $6,000 and subsequent repayments reduce by $200 each year. The repayments were calculated using an effective rate of interest of 9% per annum.

(a) Calculate the original amount of the loan.

(5 marks)

(b) Immediately after the ninth payment of interest and capital, the interest rate on the outstanding loan is reduced to 7% per annum effective. Calculate the amount of the tenth payment if subsequent payments continue to be reduced by $200 each year, and the loan is to be repaid by the original date, i.e. 20 years from commencement.

(10 marks)

(Total 15 marks)

PLEASE TURN OVER

6. To expand his business, Happy took a loan of $1million from a bank on 1 January 2020 which he will need to completely payoff by 31 December 2034.

The bank will charge interest as follows.

- 5% per annum effective for the first 5 years,
- 4% per quarter-year effective for the next 5 years, and
- 1% per month effective thereafter.

Happy will make repayments in dollars to the bank as follows.

- No repayments to be made in the first 5 years.
- In calendar years 2025-2029 he will make quarterly repayments. The repayments in 2025 will be Y per quarter, and in each subsequent year the repayments will increase by 10% meaning that in 2026 the repayments will be $1.1Y$ per quarter and so on utill in year 2029 the quarterly repayments will be 1.1^4Y.
- In calendar years 2030-2034 he will make monthly repayments. The repayments in 2030 will be $0.5Y$ per month, and in each subsequent year the repayments will increase by 5% meaning that in 2031 the repayments will be $1.05(0.5Y)$ per month and so on utill in year 2034 the monthly repayments will be $1.05^4(0.5Y)$.

(a) Show that the value on 1 January 2030 of the repayments made in the calendar years 2030-2034 is 24.5537Y.

(6 marks)

(b) Show that the value on 1 January 2025 of the repayments made in the calendar years 2025-2029 is 16.1075Y.

(5 marks)

(c) Hence, calculate the value of Y to the nearest integer.

(3 marks)

(Total 14 marks)

END OF PAPER